

1. Administrative & Study Identification

Full Study Title: A Randomized, Open-label, Inferentially Seamless Phase 2/3 Study of Izalontamab Brengitecan (BMS-986507) Versus Treatment of Physician's Choice in Patients With Previously Untreated, Locally Advanced, Recurrent Inoperable, or Metastatic Triple-negative Breast Cancer (TNBC) or ER-low, HER2-negative BC who are Ineligible for Anti-PD1/PD-L1 Treatment **Brief Study Title:** A Study of Izalontamab Brengitecan Versus Chemotherapy in Participants With Previously Untreated, Locally Advanced, Recurrent Inoperable, or Metastatic Triple-negative Breast Cancer Ineligible for Anti-PD(L)1 Drugs (IZABRIGHT-Breast01) **Short Title/Acronym:** IZABRIGHT-Breast01 **Protocol Number:** CA244-0008 **NCT Number:** NCT06926868 **Trial Status:** Recruiting **Sponsor Name:** Bristol Myers Squibb (Collaborating Sponsor: SystImmune) **Investigational Product:** Izalontamab Brengitecan (iza-bren / BMS-986507), a bispecific EGFR x HER3 antibody-drug conjugate (ADC) **Phase of Development:** Phase 2 / Phase 3 **Medical Monitor:** Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of iza-bren versus the Treatment of Physician's Choice (TPC) in patients with previously untreated, locally advanced, recurrent inoperable, or metastatic TNBC or ER-low, HER2-negative breast cancer. **Secondary Objectives:** To evaluate Overall Survival (OS) as the key secondary endpoint, Recommended Phase 3 Dose (RP3D), adverse events, objective response rate (ORR), disease control rate (DCR), duration of response (DOR), time to response, investigator-assessed PFS, safety, and health-related quality of life.

2.2 Background & Rationale There remains a significant unmet medical need for patients with metastatic TNBC or ER-low, HER2-negative breast cancer who are not eligible for anti-PD(L)1 immunotherapy or endocrine therapies. Iza-bren targets EGFR and HER3—two receptors highly expressed in most epithelial tumors—offering a novel bispecific ADC approach to deliver a topoisomerase 1 inhibitor directly to cancer cells to improve survival outcomes compared to standard chemotherapy.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Treatment of Physician's Choice: paclitaxel, nab-paclitaxel, carboplatin plus gemcitabine [Gemzar, Infugem], or capecitabine) **Blinding:** Open-label **Randomization:** Randomized (inferentially seamless allocation to iza-bren arms vs. chemotherapy arm)

3.2 Planned Enrollment & Duration Target Enrollment: 500 estimated patients **Number of Sites:** Approximately 292 locations globally **Estimated Study Start:** 11-Sep-2025 **Estimated Primary Completion:** 29-Mar-2026 (End date: 13-Mar-2028)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years old. 2. Histologically or cytologically confirmed and documented locally-advanced, recurrent inoperable, or metastatic TNBC (ER < 1%, PgR < 1%, HER2 IHC 0, 1+, or 2+ with FISH negative for HER2 gene amplification) or ER-low, HER2-negative BC (ER and/or PgR 1% to 10%, HER2 IHC 0, 1+, or 2+ with FISH negative for HER2 gene amplification) per ASCO/CAP criteria. 3. Patients with recurrent disease must have experienced disease relapse at least 6 months after finishing their last therapy with curative intent. 4. Measurable disease by CT or MRI as per RECIST v1.1. 5. Patients with TNBC must be considered ineligible for 1L chemotherapy combination treatment with an anti-PD-1 or an anti-PD-L1 due to either one of the following criteria: Investigator-determined ineligibility based on PD-L1 negative disease determined and documented prior to trial screening; experienced disease relapse between 6 to 12 months after the completion of (neo)adjuvant therapy with an anti-PD(L)1; or has a severe auto-immune disease or other contraindication.

4.2 Exclusion Criteria 1. Patients with ER-low, HER2-negative BC who are eligible, in the opinion of the Investigator, for endocrine therapy-based treatments. 2. Previous systemic therapy in the locally advanced, recurrent inoperable, or metastatic (incurable) setting. 3. Uncontrolled or severe active illnesses that contraindicate study participation. 4. Other protocol-defined Inclusion/Exclusion criteria apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Izalontamab brengitecan (iza-bren) administered on Day 1 of each 21-day cycle (Q3W dosing schedule) or corresponding dose-optimized arm. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** Continues until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for chemotherapy-related toxicities (e.g., antiemetics, growth factors for neutropenia). **Prohibited:** Other concurrent anticancer therapies, investigational agents, or immunotherapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Biopsy/Pathology Review, Baseline Imaging (CT/MRI)
Baseline	Day 0	Randomization, First IV Dose
Interim	Every 3 Weeks	Treatment administration, Safety Labs, Efficacy Assessment (Scans every 6-9 weeks)
End of Study	30 Days post-treatment	Final Vital Signs, Exit Interview, Safety Check, Transition to Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS)
Measurement Method: Assessed using Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1) by blinded independent central review (BICR).

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) * Recommended Phase 3 Dose (RP3D) of BMS-986507 * Number of participants with treatment-related Adverse Events (AEs) * Investigator-assessed PFS, Objective Response Rate (ORR), and Disease Control Rate (DCR) * Duration of Response (DOR), time to response, and health-related quality of life (HRQoL)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via regular clinical visits before each dosing cycle, recording the number of participants with treatment-related Adverse Events (AEs). Particular focus on hematologic AEs (e.g., anemia, leukopenia, neutropenia) commonly associated with ADCs carrying a topoisomerase 1 inhibitor. **Serious Adverse Events (SAEs):** Routine expedited reporting timelines (typically 24 hours). **Data Monitoring Committee (DMC):** An independent committee will review safety and efficacy data, particularly given the seamless Phase 2 to Phase 3 design.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy endpoints (PFS, OS, ORR); Safety Set for all subjects

who receive at least one dose of study treatment. **Sample Size Justification:** Target enrollment of 500 participants, powered sufficiently to test the superiority of iza-bren over the physician's choice chemotherapy in delaying progression. **Handling of Missing Data:** Time-to-event outcomes (PFS, OS) will use Kaplan-Meier methodologies and appropriate censoring rules for patients lost to follow-up or without documented progression.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based monitoring incorporating both on-site visits and centralized remote data review of EDC entries. **Audit Trail:** Continuous tracking via an electronic data capture (EDC) system with data clarification mechanisms and validated BICR imaging uploads.

11. Study Identifiers & Governance

Primary ID: CA244-0008 **Registry Number:** NCT06926868 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study of Izalontamab Brengitecan Versus Chemotherapy in Participants With Previously Untreated, Locally Advanced, Recurrent Inoperable, or Metastatic Triple-negative Breast Cancer Ineligible for Anti-PD(L)1 Drugs (IZABRIGHT-Breast01) **Official Regulatory Title:** A Randomized, Open-label, Inferentially Seamless Phase 2/3 Study of Izalontamab Brengitecan (BMS-986507) Versus Treatment of Physician's Choice in Patients With Previously Untreated, Locally Advanced, Recurrent Inoperable, or Metastatic Triple-negative Breast Cancer (TNBC) or ER-low, HER2-negative BC who are Ineligible for Anti-PD1/PD-L1 Treatment

12. Clinical Framework

Primary Disease Areas: Breast Neoplasms (Triple-Negative Breast Cancer and Mammary Neoplasms) **Diagnosis Threshold:** HER2 IHC 0, 1+, or 2+ with FISH negative for HER2 gene amplification; ER/PgR < 1% (TNBC) or ER/PgR 1-10% (ER-low) **Medical Methodology:** Bispecific Antibody-Drug Conjugate (ADC) vs. standard chemotherapy **Optimization Target:** PFS prolongation and cellular toxicity via Topoisomerase 1 inhibition directed by EGFR/HER3 targeting.

13. Study Procedures & Methodology

Management Pathway: Standard IV infusion pathway for systemic oncology therapy. **Education Protocol (HCP):** Site initiation

